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SUPERSEDING
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MILITARY SPECIFICATION

TRAILERS, TANK: POTABLE WATER AND GENERAL PURPOSE, 500 GALLON, TWO AND FOUR-WHEEL, MODIFIED COMMERCIAL

This specification is approved for use by all departments and agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers 500 gallon capacity, two-wheel and four-wheel (wagon type) tank trailers for hauling and dispensing liquids (see 6.1 and 6.3). Vehicles procured under this specification are commercial items which are warranted by the manufacturer as specified in procurement documents.

1.2 Classification. The trailer shall be one of the types, sizes and models shown in table I, as specified (see 6.2).

* TABLE I. Trailer classification.

Type	Nomenclature	Size	Model
I	Potable water	500	2W
II	General purpose (water, diesel fuel and oil)	500 500	2W 4W

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Tank-Automotive Command, ATTN: AMSTA-UED, Warren, MI 48397-5000, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A

FSC-2330

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1.2.1 Size. The size of the trailer shall be determined by tank capacity as follows:

Size 500 - 500 gallon

1.2.2 Model. The model of the trailer shall be determined by the number of wheels as follows:

Model 2W - Two-wheel
Model 4W - Four wheel (wagon-type)

2. APPLICABLE DOCUMENTS

2.1 Government documents.

* 2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS
FEDERAL

*	A-A-393	- Extinguishers, Fire, Dry Chemical (Portable).
	ZZ-H-561	- Hose, Rubber, and Hose Assemblies, Rubber, Smooth Bore, Water Suction and Discharge.

MILITARY

MIL-C-83413	- Connector, Electrical Ground.

STANDARDS
FEDERAL

FED-STD-297	- Rustproofing of Commercial (Nontactical) Vehicles.

MILITARY

MIL-STD-1223	- Non-Tactical Wheeled Vehicles Treatment, Painting, Identification Marking, and Data Plate Standards.
MS 51335	- Pintle Assembly, Towing - 18,000 Lbs. Capacity, Manual Release.
MS 51336	- Lunette - Coupler, Drawbar, Ring.
MS 51339	- Coupler, Drawbar, Ring-Offset.

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* (Unless otherwise indicated, copies of federal and military specifications, standards and handbooks are available from the Naval Publications and Forms Center, Military Specifications and Standards, Bldg. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094).

* 2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein. Unless otherwise specified the issues are those cited in the solicitation.

* DEPARTMENT OF DEFENSE

Department of Defense Index of Specifications and Standards (DODISS).

(Copies of the DODISS are available on a yearly subscription basis either from the Government Printing Office for hard copy, or microfiche copies are available from the Director, Navy Publication and Printing Service Office, 700 Robbins Avenue, Philadelphia, PA 19111-5093.)

DEPARTMENT OF TRANSPORTATION (DOT)

Federal Motor Carrier Safety Regulations.
Federal Motor Vehicle Safety Standards.
Regulations Governing Transportation of Dangerous Articles in Tank Motor Vehicles.

(Application for copies of DOT publications should reference the Code of Federal Regulations, 49 CFR, and the Federal Register, and should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402.)

FOOD AND DRUG ADMINISTRATION (FDA)

Indirect Food Additives: Polymers (21 CFR 177)

(Application for copies of FDA publications should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402.)

* 2.2. Non-Government publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN NATIONAL STANDARD INSTITUTE, INC. (ANSI)

American National Standard Pipe Thread Series.

(Application for copies of ANSI publications should be addressed to the American National Standard Institute, 1430 Broadway, New York, NY 10018.)

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* THE EUROPEAN TYRE AND RIM TECHNICAL ORGANISATION (ETRTO)
Standards Manual

(Application for copies of the ETRTO publications should be addressed to the European Tyre and Rim Technical Organisation, 32, Avenue Brugmann, 1060 Brussels, Belgium.)

SAE, INC.
SAE Standards and Recommended Practices

	J318	- Air Brake Gladhand Service (Control) and Emergency (Supply) Line Couplers - Trucks, Truck-Tractors, and Trailers (DoD adopted).
*	J350	- Spark Arrester Test Procedure for Medium Size Engines.
	J551	- Performance Levels and Methods of Measurement of Electromagnetic Radiation from Vehicles and Devices (30-1000 MHz).
	J560	- Seven-Conductor Electrical Connector for Truck-Trailer Jumper Cable.
	J588	- Turn Signal Lamps.
	J682	- Rear Wheel Splash and Stone Throw Protection.
	J1067	- Seven-Conductor Jacketed Cable for Truck-Trailer Connections.

(Application for copies of SAE publications should be addressed to SAE, Inc., 400 Commonwealth Drive, Warrendale, PA 15096.)

THE TIRE AND RIM ASSOCIATION, INC.
Year Book.

* (Application for copies of Tire and Rim Association publications should be addressed to The Tire and Rim Association, Inc., 175 Montrose West Ave., Copley, OH 44321.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

* 2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document shall take precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 Standard vehicle and accessories. Except as specified in 3.1.1 through 3.1.1.4, the vehicle, components assemblies, and accessories to be delivered under the contract shall be standard or optional items, which meet or exceed the requirements of this specification. All items shall be as represented in the trailer manufacturer's technical data book. Special or mounted equipment shall be as represented and rated in the body and equipment manufacturer's technical data. Technical data shall be limited to specifications and technical material identical to that furnished to the authorized company representatives for selection of vehicle models and components, and shall be available to the engineering offices of the procuring activity, prior to delivery of the items.

3.1.1 Special requirements. In addition to the standard vehicle and components specified in 3.1, the vehicle shall be furnished with special equipment as specified herein.

* 3.1.1.1 Treatment and painting. The vehicle shall be treated and painted in accordance with MIL-STD-1223. As specified by the procuring activity for the appropriate military service (see 6.2), the exterior color shall be in accordance with MIL-STD-1223.

* 3.1.1.2 Marking and data plates. As specified by the procuring activity for the appropriate military service (see 6.2), identification marking and data plates shall be in accordance with MIL-STD-1223.

* 3.1.1.3 Rustproofing. When specified (see 6.2), the vehicle shall be rustproofed in accordance with FED-STD-297. Stainless steel surfaces need not be rustproofed.

3.1.1.4 Toxic products and formulations (type I). The materials used in the construction or treatment of surfaces on type I trailers, which may be in contact with potable water during use, shall have no adverse effect on the health of personnel. Materials used shall be non-toxic, relatively non-absorbent, resistant to fat and normal cleaning with bactericidal solutions, and shall not contaminate the water or affect the smell or taste of the water. Nozzle and faucet seals shall be free of asbestos impregnated materials. Parts, fittings, pumps and piping in contact with water shall be of corrosion resistant material not affecting the potability of the water. Rubber or plastic materials used on the interior surfaces of gaskets or hoses that contact potable water shall meet the requirements of 21 CFR part 177, except materials constructed of acrylonitrile and co-polymers shall not be used. Manufacturers shall submit evidence of conformance to the requirements of this paragraph for review and approval by the procuring activity. Such evidence shall be submitted within 10 days from date of receipt of contract and shall identify all materials that come in contact with potable water. Materials shall withstand handling and usage in ambient temperatures of minus 40 degrees Fahrenheit (^{0}F) to plus 135^{0}F.

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* 3.2 General Design.

* 3.2.1 Federal Motor Vehicle Safety Standards. The vehicle shall comply with all Federal Motor Vehicle Safety Standards applicable to the type of trailer furnished and in effect on the date of manufacturer.

* 3.2.2 Dissimilar Metals. All dissimilar metals used throughout the vehicle shall be insulated from one another to prevent galvanic or electrolytic action.

* 3.2.3 Ratings. Vehicle rating shall be the manufacturer's published ratings. Component and vehicular ratings shall not be raised to meet the requirements of this specification. When published ratings are not available, verification of rating shall be available to the appropriate engineering office of the procuring activity.

* 3.2.4 Accessibility. The design of the vehicle and optional equipment shall permit access for routine servicing and shall permit access for replacement and adjustment of component parts and accessories with minimal disturbance of other components and systems.

* 3.2.5 Prohibited materials. Asbestos materials shall not be used in any form in any part of the vehicle. No item, part or assembly shall contain radioactive materials in which the specific activity is greater than 0.002 microcurie per gram or activity per item equals or exceeds 0.01 mircocuries.

3.3 Performance. Vehicle performance as specified in 3.3.1 through 3.3.5 shall be met by the trailer when fully equipped, completely serviced, loaded with the rated payload and towed by a towing vehicle over improved and unimproved, hard uneven terrain. The vehicle shall show no evidence of failure or malfunction of trailer, components, attachments and accessories.

3.3.1 Towing speed and trailing ability. The trailer shall follow the towing vehicle over dry, level, smooth, prepared hard surfaced roads at sustained speeds up to 60 miles per hour (mph), and over dry, unimproved and gravel roads at speeds up to 25 mph. The trailer shall follow the towing vehicle without exceeding the tracking deviations specified in Federal Motor Carrier Safety Regulation 393.70(a).

* 3.3.2 Turning ability. When coupled to a towing vehicle operating in its minimum turning circle, the trailer shall follow without cramping. The trailer shall be capable of moving through an angle of not less than 60 degrees in relation to a 102-inch wide towing vehicle without interference between the two vehicles. The angle shall be measured between the longitudinal axis of the trailer and the longitudinal axis of the towing vehicle at the lunette.

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3.3.3 Suspension system. The axle, springs, accessories and components of the suspension system shall withstand the imposed loads without evidence of permanent set, part failure or malfunctioning under the operating conditions specified.

3.3.4 Service brakes. The service brakes of the trailer in conjunction with the towing vehicle (truck-trailer combination), under all conditions of loading, shall control, decelerate, and stop the vehicle combination in accordance with Federal Motor Carrier Safety Regulation 393.52.

3.3.5 Parking brakes. The parking brakes shall hold the trailer with the specified payload on a dry, hard surfaced grade of not less than 30 percent when uncoupled from the towing vehicle, resting on the running gear (for model 4W), or on the landing leg and running gear (for model 2W), and headed up and down the incline. The parking brake lever shall have sufficient moment arm to require not more than 30 pounds of effort for application and release.

3.4 Pay1oad and weight distribution.

3.4.1 Payload. The rated payload capacity shall consist of 500 gallons. When calculated loads are required, water weight shall be computed using 8.3 pounds per gallon.

3.4.2 Weight distribution. For model 2W, with the longitudinal axis of the trailer horizontal, and with the trailer unloaded or loaded with the specified payload, the center of gravity of the trailer shall be forward of the axle. The maximum vertical load on the lunette shall be 250 pounds with the trailer empty and 650 pounds with the trailer loaded with the specified payload. For model 4W, weight distribution shall be such that there are approximately equal loads at the ground, on each wheel, when loaded with the specified payload.

3.5 Trailer components.

3.5.1 Frame. The trailer shall be equipped with a frame of all-steel welded construction, with each main frame member in one piece without splices or joints. A sufficient number of full width crossmembers shall be furnished to assure stability with the specified payload. The sprung weight of the loaded vehicle shall not impose a fiber stress greater than 50 percent of the yield strength of the materials used under the mobile operating conditions specified. All parts of the chassis frame, where welding is required, shall be electrically welded.

* 3.5.1.1 Drawbar and coupling device (model 2W). The model 2W trailer shall be equipped with a drawbar of the "A" frame type formed by the continuation of the trailer main frame side rails. The coupling device shall be a standard lunette ring conforming to MS 51336. The

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coupling device shall be installed and positioned for attachment to the towing vehicle in such a manner that it shall have a 4-inch vertical range of adjustment. The lunette shall be positioned so that the longitudinal axis of the level tank is parallel to the lunette. With the trailer fully loaded, level and resting on the landing wheel, and with the lunette in its lowest position, the height of the level lunette shall be 30 inches, plus or minus 1 inch, and at its highest position, the height of the lunette shall be 34 inches, plus or minus 1 inch.

3.5.1.2 Landing leg (model 2W). Model 2W trailer shall have a retractable landing leg with caster type wheel and a permanently installed swivel type cranking handle. The landing leg and its related parts shall withstand, without permanent deformation, the combined static and dynamic forces due to coupling and uncoupling operations. Angular landing leg braces, extending down to not more than 16 inches above ground level, shall be provided. The landing leg shall operate freely and shall be protected from malfunctioning caused by foreign matter. The landing wheel, and any rotating components of the landing leg, shall be fitted with antifriction bearings. When placed in travel position, the landing leg shall remain positively locked and shall provide not less than 15 inches of ground clearance in the course of normal towing operations.

3.5.1.3 Rear support legs (model 2W). Two support legs shall be furnished on model 2W, one on each side at the rear of the trailer, to rigidly support the trailer and to prevent backward tilting of the trailer when the trailer is uncoupled from the towing vehicle. The legs shall be adjustable, and shall have self contained provisions for locking in both travel and support positions. The legs shall provide a ground clearance of not less than 15 inches when in travel position.

3.5.1.4 Tow bar and steering (model 4W). Model 4W trailer shall be furnished with a tow bar conforming to Federal Motor Carrier Safety Regulation 393.70(c) and (d). The trailer shall be equipped with a standard lunette ring conforming to MS 51336. The coupling device shall also be of the offset type conforming to MS 51339 or otherwise adjustable to provide a 4-inch vertical range of adjustment. The design of the trailer shall incorporate an Ackerman steering system with steering capability of at least 35 degrees to the right and the left of the vehicle centerline. A non-ferrous plate, secured with non-ferrous screws, and marked "DO NOT BACK UP" shall be provided on the front of the trailer in a readily visible location. The tow bar shall be hinged, and shall be capable of being placed in a vertical position. Means shall be provided to retain the towbar in the vertical position. Lubrication fittings shall be furnished. The lunette shall be located 28 inches plus or minus 1 inch above ground level, with the tow bar parallel to the ground surface, the tank level and the coupling device in its lowest position of adjustment.

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3.5.1.5 Safety chains. The trailer shall be equipped with safety chains. Two chains with eyes shall be provided. The chains shall conform to Federal Motor Carrier Safety Regulation 393.70(d).

3.5.2 Suspension. The trailer shall be equipped with springs of the semi-elliptical type. The suspension system shall have a rated capacity at least equal to the load imposed on each member, measured at the ground, with the trailer loaded as specified in 3.4.1.

3.5.3 Axles. Model 2W trailers shall have a single axle. Model 4W trailers shall have two axles. Axles shall have a rated capacity at least equal to the load imposed on each axle, measured at the ground, with the trailer loaded as specified in 3.4.1 and 3.4.2.

* 3.5.4 Wheels, rims, tires and tubes. Model 2W trailer shall be equipped with single rear wheels and tires. Model 4W trailer shall be equipped with single front and single rear wheels and tires. The rim size and tire ratings shall conform to Tire and Rim Association or European Tyre and Rim Technical Organisation recommendations for the type and size of tires furnished. Tire size, tire load range (ply rating), and rim sizes shall be the same for all wheels on the vehicle. Multi-piece rims shall not be furnished.

* 3.5.4.1 Tires. Tires shall be of the tubeless type with highway tread. Unless otherwise specified, low profile or standard profile tires may be furnished. When specified (see 6.2), low profile tires shall be furnished. Tires shall be of the steel belted radial type. Tires shall be of rated capacity at least equal to the load imposed on each tire, measured at each wheel, at the ground, with the trailer loaded as specified in 3.4.1 and 3.4.2. Tires shall conform to Tire and Rim Association or to European Tyre and Rim Technical Organisation recommendations.

3.5.4.2 Inner tubes. When tube-type tires are furnished, inner tubes shall be of heavy duty type, and of the proper size for the tires furnished. Tubes for radial tires shall be of the radial tire tube type. Tire flaps shall be provided for tube-type tires in accordance with Tire and Rim Association recommendations.

* 3.5.4.3 Spare wheel or rim. When specified (see 6.2), a spare wheel or rim shall be mounted on a carrier and installed in a readily accessible location.

3.5.4.4 Spare tire assembly. When specified (see 6.2), a spare tire assembly shall be furnished. The spare tire assembly shall include an inflated spare tire mounted on the spare wheel or rim. The spare tire shall be of the same size, tread design and load range as the tires in use on the vehicle.

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* 3.5.4.5 Spare tire carrier. When specified (see 6.2), a spare tire carrier shall be furnished. The carrier shall enable the safe removal and installation of the spare tire assembly from and to the vehicle and carrier without personnel positioning themselves or any part of their body under the spare tire assembly. Threaded fasteners, when used to secure the spare in the carrier, shall be constructed of or plated with corrosion-resistant material.

3.5.5 Brakes.

* 3.5.5.1 Service brakes. Unless otherwise specified (see 3.5.5.2), the trailer shall be equipped with a 12-volt electric brake system wired in accordance with 3.5.6.3. The system shall include a manual controller kit for mounting on the dash of the towing vehicle, a controller kit for synchronizing the trailer electric brakes with the brakes of the towing vehicle and an emergency breakaway kit. Kits shall be complete with all necessary parts (except storage battery for breakaway kit) and installation instructions. Unless otherwise specified, the synchronizing controller shall be for use with hydraulic brake equipped towing vehicles. When specified (see 6.2), the synchronizing controller shall be for use with air brake equipped towing vehicles. The kits shall be boxed, marked as to contents and securely attached to the trailer for shipment. The brake linings shall be of nonasbestos material.

3.5.5.2 Air-hydraulic brakes. When specified (see 6.2), the trailer shall be equipped with air-hydraulic type service brakes in lieu of electric brakes. Hose connections shall be in accordance with SAE Standards and Recommended Practices. The contractor shall maintain in his file a specification for adjustment of clearances between the brake lining and the brake drum for new and already applied brakes. The adjustments for the service brakes shall allow sufficient clearance between the brake shoe lining and the brake drum to prevent drag when in released position and to maintain a solid grip without slipping when in applied position and shall be capable of preventing the wheels from turning with the specified payload.

3.5.5.3 Brake system. The stabilization of the air system at the rated working pressure shall be accomplished with the hydraulic system free of air and with both air lines connected to a compressed air source. An air pressure gage shall indicate that when the brakes are released, the pressure drop shall not exceed 1 psi per minute, nor shall it exceed 2 psi per minute with the brakes normally applied. In the intervening motion, there shall be no hydraulic fluid leakage or air leakage. The following equipment shall be furnished as a minimum:

(a) Standard breakaway features.
(b) Air reservoir having not less than 1000 cubic inch capacity

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(c) Bleed-down relay emergency valve
(d) Hydraulic master cylinder with air chamber
(e) Air connectors with SAE J318 gladhand couplers mounted at front of the trailer
(f) Dummy gladhand couplers with retaining chains or cables
(g) Permanent identification tags for emergency and service lines.

3.5.5.4 Parking brake. Manually applied parking brakes shall be furnished. The parking brakes shall conform to Federal Motor Carrier Safety Regulation 393.41. When brake cables are used, the cables shall be of corrosion resistant steel. Provisions shall be made for slack adjustment. Lubrication fittings shall be provided.

3.5.5.5 Wheel chocks. When specified (see 6.2), four wheel chocks shall be provided. The wheel chocks shall be secured to the trailer with corrosion resistant chains or cables. Provisions for stowage of each wheel chock near the wheel to be chocked shall be provided.

3.5.6 Electrical system.

* 3.5.6.1 Lighting. The electrical lighting system shall be 12 volts. All vehicle lights, reflectors, and wiring (including electric brakes) shall conform to Federal Motor Carrier Safety Regulations 393.14, 393.19, 393.20, 393.22, 393.23, 393.25, 393.26 (a, b, c and d), and 393.27 through 393.33. At least one pair of brake lights shall override the four-way emergency flasher or the two systems shall be independent of each other. Lights and reflectors shall not be mounted on vehicle bumpers.

3.5.6.2 Electrical receptacle. A trailer electrical receptacle with a spring-loaded cover assembly shall be installed on the forward end of the drawbar. The receptacle shall conform to SAE J560, with conductors connected and color-coded as specified in SAE J560.

3.5.6.3 Brake wiring. Electric brakes shall be wired to the auxiliary circuit of the SAE J560 seven conductor electrical connector. All wiring shall be properly mounted and secured with clamps. Wiring shall be heavy wall thermoplastic insulated cable and shall be enclosed in conduit.

* 3.5.6.4 Electrical jumper cable. Unless otherwise specified (see 6.2), an electrical jumper cable conforming to SAE J1067, with a cable plug at each end conforming to SAE J560, shall be provided. Both plugs shall be equipped with a grip for withdrawing from the connector sockets. Stowage provisions with means of hanging the cable with plugs pointing down shall be provided. The cable shall be not less than 3 feet longer than the distance from the trailer electrical receptacle to the lunette. The cable shall be boxed, marked and secured to the vehicle for shipment.

* 3.5.7 Bumper. A rear bumper shall be furnished and shall conform to Federal Motor Carrier Safety Regulation 393.86 and DOT Safety Regulation 178.345. The bumper shall extend not less

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than 3 inches beyond the rear of the tank. When a pintle is furnished, the bumper shall be constructed to allow for installation of a pintle assembly.

* 3.5.8 Fenders. The trailer shall have fenders over each wheel. The fenders shall prevent mud and water from splashing on the tank. The top of the fenders shall be essentially flat. The top of the fenders shall be fabricated of diamond tread steel or covered with abrasive-type paint (ships grip). The top of the fenders shall be capable of rigidly supporting a 200 pound servicing personnel standing on the top. The fenders shall be rigidly attached to the chassis frame. Rubber mudflaps shall be installed to the rear of the wheels and shall conform to SAE J682. A metal strip not less than 0.125 inch thick and not less than 1 inch wide, extending the entire width of the mud flap, shall be installed to prevent the bolt heads or bolt nuts from damaging the mud flap. As an alternate method of attaching the mud flaps, tabs or clips with minimum surface contact dimensions of 1 inch high by 1.250 inch wide by 0.094 inch thick shall be furnished at each bolt. Access steps and grab handles, to assist personnel in climbing on top of the fenders, shall be provided.

3.5.9 Pintle. When specified (see 6.2), a pintle assembly conforming to MS 51335-2 (solid type) shall be installed on an adequately supported device at the rear of the chassis frame. The rearmost portion of the pintle assembly shall project beyond the rear of the vehicle. Two trailer safety chain attachment devices, one adjacent to each side of the pintle, shall be provided. Each attachment device shall provide an ultimate strength of not less than 18,000 pounds and shall be capable of accommodating a standard grab hook for a 0.375-inch chain (2.813 wide, 0.734 thick, and with a 0.484 throat diameter).

3.5.10 Lubrication. All moving parts of the equipment normally requiring lubrication shall be provided with suitable means for such lubrication. The lubrication fittings shall be standard commercial type with a surface-check ball.

3.5.11 Dimensions and clearances. The trailer, when uncoupled from the towing vehicle and without payload, on level ground, shall conform to the dimensions specified in table II.

TABLE II. Trailer dimensions (inches).

Characteristic	Dimension
Overall width (maximum)	96
Overall height (maximum)	68
Overall length (without engine driven pump) (maximum)	168
Overall length (with engine driven pump) (maximum)	200
Ground clearance (minimum)	15

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3.6 Tank body and equipment. The trailer shall be designed and constructed to meet the requirements of DOT Safety Regulation 178.345 for material, structural integrity, joints, supports and anchoring, circumferential reinforcement and minimum thickness, except in no case shall the minimum thickness of any part of the tank be less than 12 gage (0.1046 inch). Minimum thickness requirements shall be based on a product density of 8.3 pounds per gallon. The type II trailer shall be designed and constructed for the legal transportation of liquid hazardous materials by interstate carriers. Accordingly, the vehicle shall conform in all respects to the regulations of DOT governing the transportation of hazardous materials in tank motor vehicles. Conformance to DOT Safety Regulation 178.345 and DOT Safety Regulation 178.346, as published in the latest issues of the Code of Federal Regulations and as modified by all issues of the Federal Register published and in effect prior to completion of the vehicle, shall be total and without exception.

3.6.1 Material. The tank shell, baffles, bulkheads, pads, manholes and outlets shall be of type 304 stainless steel. The thickness of the shell, head and baffle sheets shall be not less than 12 gage (0.1046 inch) stainless steel. Tank combinations of mild steel, high strength low alloy steel and austenitic stainless steel, as permitted by DOT Safety Regulation 178.345, shall not be furnished. All piping, pipe fittings, valves, faucets, vents, strainers, hose fittings, reducers, hose adapters, dust caps, dust plugs, and dust cap and dust plug security chains (a) shall be of stainless steel (brass faucets may be furnished) and shall conform to 3.1.1.4 for type I or (b) shall be constructed of or plated with corrosion-resistant material for type II. Where dissimilar metals are permitted and used, effective means shall be taken to impede electrolytic action. Teflon packing and tape shall not be used on or adjacent to parts with an aluminum content.

3.6.2 Capacity. The liquid capacity of the tank shall be not less than 500 gallons, plus at least 3 percent expansion space by volume. The tank shall have a single compartment.

3.6.3 Tank configuration. The tank shall have an elliptical, circular or modified rectangular or modified rectangular section.

* 3.6.4 Baffles. Where the provisions of DOT Safety Regulation 178.345 would require circumferential reinforcements., baffles shall be used in lieu of ring stiffeners. Baffles shall be flanged and dished; or flat, reinforced and flanged. A half round opening of not less than 3.375 inch radius shall be provided at the bottom of each baffle. Openings of equivalent size shall be provided at the top of the baffles and at the 3 o'clock and 9 o'clock positions in the baffles. Each baffle shall also be provided with not less than an 18-inch diameter manway opening. The edges of the manway openings shall have a flange of approximately 1 inch.

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3.6.5 Tank support. Full length box-type, V-type or Z-type longitudinal subframe members shall be furnished. Full length welding pads of not less than 12 gage (0.1046 inch) type 304 stainless steel shall extend not less than 1 inch beyond the flange on the longitudinal subframe members and shall be continuously welded to the tank proper.

3.6.5.1 Box-type longitudinals. Box-type longitudinal subframe members shall be fabricated of not less than 12 gage (0.1046 inch) mild steel. A flange of not less than 1 inch wide shall be provided at the top of both the outer and inner vertical legs of the boxes. Flanges shall be continuously welded to the pads the full length of the tank.

3.6.5.2 V-type 1ongitudinals. V-type longitudinal subframe members shall be fabricated with not less than 7 gage (0.1793 inch) mild steel inner and outer legs. A flange not less than 1 inch wide shall be provided at the top of both the inner and outer legs. Flanges shall be continuously welded to the pads the full length of the tank.

3.6.5.3 Z-type longitudinals. Z-type longitudinal subframe members shall be fabricated entirely of not less than 7 gage (0.1793 inch) mild steel. The width of the upper Z member shall be not less than 2 inches and shall be continuously welded to the pads the full length of the tank along both sides of the Z. With the Z-type construction, additional type 304 stainless steel pads shall be continuously welded to the tank proper at the front and rear and at each baffle. Bolsters of 0.188 inch mild steel shall be properly integrated with the Z-type longitudinals and continuously welded to the pads at the front and rear and at each baffle.

* 3.6.6 Manhole. The tank shall be equipped with a round or oval manhole opening not less than 19 inches in major diameter. The manhole sides and lip shall be constructed of stainless steel having a carbon content of not more than type 304 sheet steel. A light weight, liquid-tight, easy opening, self-latching manhole cover, or a bolted cover with a liquid-tight, easy opening, self-latching fill cover not less than 10 inches in diameter, shall be supplied. All metal parts of the cover exposed to contact with liquids in the tank shall be fabricated of type 304 stainless steel. Hinged covers shall be hinged at the forward edge. The manhole shall be located longitudinally on the tank such that the center of the manhole shall be forward of the center of the tank and shall be accessible from the top of the (front) fenders.

3.6.7 Capacity marker (type II). On type II trailer, a capacity marker or indicator disc of corrosion resistant construction shall be installed at the side of the manhole.

3.6.8 Vents. Air vents shall be provided to allow for expansion of the specified liquid in the tank and for discharging the tank at maximum rate with all faucets open and the manhole and fill cover sealed. Vents shall be adequate to limit the tank pressure and vacuum under these

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conditions to those permitted by DOT Safety Regulation 178.346. Vents shall remain sealed When not fulfilling their purpose of equalizing internal and external pressures. The vents shall be easily removable for cleaning. Normal and emergency venting in accordance with DOT Safety Regulation 378.346, shall be provided on type II.

3.6.9 Mounting. The tank longitudinal subframe members shall be secured to the chassis frame longitudinals with U-bolts, twin studs, or by brackets, and shall include wood breaker strips.

3.6.9.1 U-bolts or twin studs. When U-bolts or twin studs are used, there shall be not less than four U-bolts or twin studs per side, each having 0.562 inch minimum body diameter with 0.625 inch minimum thread diameter. Tie-plates shall be at least 0.500 inch thick. Vehicle chassis frame shall be braced, using blocks of wood at each mounting point unless the mounting point is located at a full depth, frame crossmember. Blocks shall incorporate a keeper strap or groove for a mounting bolt, and shall be of a width to assure retention. Two tie-back straps shall be provided, one on each side of the rear portion of body subframe, to maintain body alinement on the vehicle chassis

3.6.9.2 Brackets. When brackets are used, they shall be bolted to the web of the chassis frame rails. The body mounting brackets shall provide a means of drawing down the body on the chassis rails and provision shall be made to prevent lateral shifting of the breaker strips. Attachments shall not interfere with, nor obstruct, chassis components.

3.6.9.3 Breaker strips. A wood breaker strip 1 inch plus 0.063 inch, minus 0 inches finished thickness shall be installed between both of the tank longitudinal subframe members and the vehicle chassis frame. The width of each breaker strip shall be at least equal to the width of the corresponding trailer chassis frame rail.

3.7 Additional requirements for type I potable water trailer.

* 3.7.1 Piping. Discharge of the water shall be by gravity through a pipe outlet 3 inches in nominal diameter installed at the tank's lowest forward bottom section to assure maximum drainage of water. A union or flexible coupling for disconnecting the piping system from the tank outlet shall be installed as near the tank outlet as possible.

3.7.2 Main shutoff valve. A shutoff valve shall be installed in the main outlet line between the tank and a distribution manifold. The shutoff valve shall be of the quick-opening, lever-operated or butterfly gate valve type, not less than 3 inches in nominal diameter.

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3.7.3 Distribution manifold and faucets. A distribution manifold and faucets shall be installed at the front of the tank. The distribution manifold shall be not less than three inches in nominal diameter. The manifold shall be used to distribute water to the following: one 1.250 inch and two 0.500 inch faucets housed in a faucet box enclosure located on the right (curb side) end of the manifold, and one 1-inch and two 0.500 inch faucets housed on the opposite side in a second faucet box enclosure. The faucets shall be of the self-closing type with down spout outlet.

3.7.4 Faucet enclosure boxes. Two faucet enclosure boxes shall be installed, one on each side of the trailer, forward of the tank. The boxes shall be constructed of bright or satin finish stainless steel and shall be unpainted. Welds shall be ground smooth and polished. The interior shall be free of crevices. Each box shall be an integral unit consisting of four sides, a bottom and a cover. The covers shall be spring-locked both when in open and closed positions. Covers in open position shall provide for unobstructed accessibility to the faucets. The covers in closed position shall be capable of rigidly supporting a 200 pound man stepping on the covers. The design and construction shall not restrict the turning radius of the towing vehicle or project excessively beyond the "A" frame construction.

3.7.5 Work platform. A work platform for use by personnel in filling the water tank shall be furnished immediately forward of, and between, the faucet box enclosures. The platform shall be not less than 36 inches long and the width shall conform to the width of the "A" frame. The platform floor shall be constructed of steel subway grating or equal, with space between the water tank and the subway grating to prevent galvanic action of dissimilar metals.

3.7.6 Tank drain. A discharge drain shall be provided and located at the lowest rear bottom section of the tank for draining water used for cleaning and flushing the tank. The drain shall consist of a male, 1.500 inch gate valve faucet and a 1.500 inch cap on the discharge outlet of the faucet. The 1.500 inch cap shall be furnished with a security chain or cable. The gate valve handle shall be located in a position completely protected by the rear bumper and with sufficient clearance to avoid interference with operations by the bumper.

3.7.7 Outer jacket insulation. Unless otherwise specified (see 6.2), the tank shall be encircled with not less than 1.500 inch thick fiberglass insulation or foam type insulation. The insulation shall have a K factor of not more than 0.15 and a density of not less than 2.0 pounds per cubic foot. The tank insulation shall be covered with an outer jacket of type 304 stainless steel. The stainless steel jacket shall be welded on the outside and shall be waterproof. The minimum thickness of the outer jacket shall be 0.059 inch.

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3.7.8. Hand water pump. When specified (see 6.2), a manually operated rotary or piston type pump, constructed of or plated with corrosion resistant materials, shall be provided. The pump shall be located at the front of the tank and shall be used to load and unload the tank. The installation shall include a combination suction and filler pipe at the front of the tank and two hoses. The suction and filler pipe shall be not less than 3 inches in nominal diameter and shall extend from the top, outside of the tank to within 1 inch of the bottom, inside of the tank. The top of the suction and filler pipe shall be provided with a cam lock fitting and a dust plug or dust cap complete with a security chain. The hoses shall be of not less than 1.500 inch inside diameter, noncollapsible suction type. The hose shall be fitted with cam lock fittings, adapters and reducers as required, to interconnect the top of the 3-inch suction and filler pipe with both the inlet and outlet ports of the hand water pump, and to interconnect the hoses with each other. One hose shall be not less than 15 feet in length and the other shall be not less than 25 feet in length. The hoses shall be stowed on the work platform or on brackets on the side of the tank provided the overall width of the vehicle is not increased to more than 96 inches. The hand water pump shall be capable of lifting water from not less than 15 feet below ground level and discharging water not less than 15 feet above ground level through the hoses specified. Dust caps or dust plugs with security chains or cables shall be provided for each end of each hose and on both of the pump ports. The hoses and hand water pump shall conform to 3.1.1.4.

3.8 Additional requirements for type II general purpose trailer. Type II trailer shall be equipped with a tank capable of handling liquids such as water, diesel fuel and oil.

3.8.1 Emergency valve. The tank shall be provided with a 3-inch nominal diameter emergency valve. The material used in the construction of the valve shall be compatible with the liquids specified and the metal construction of the tank and its components. The valve shall be located at the lowest rear bottom section of the tank or, when a plumbing system and engine driven pump are specified (see 3.8), at the front bottom. Unless a plumbing system and engine driven pump are specified, the emergency valve control shall be located at the rear of the trailer, near the discharge faucet (see 3.9.21 for the control location when a plumbing system and engine driven pump are specified). The valve shall be of the quick opening and quick closing type with the open position controlled by fusible links to insure automatic closing in case of fire.

3.8.2 Discharge. Both discharge and draining of the tank for cleaning purposes shall be through the emergency valve with 3 inch nominal diameter piping to a 3-inch nominal diameter gate valve discharge faucet. The faucet shall have a 3-inch cam lock fitting for gravity discharge. A cam lock dust cap, complete with security chain or cable, shall be provided. Unless a plumbing system and engine driven pump are specified (see 3.9), the faucet shall be located at the rear of the tank. The installation shall be in a position completely protected by the rear bumper and with sufficient clearance to avoid interference with operations by the bumper.

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3.8.3 Discharge hose. A 3-inch nominal diameter hose for coupling to the discharge faucet specified in 3.8.2 shall be provided. The hose shall be provided with cam lock fittings. A dust cap shall be provided at the male end and a dust plug at the female end. The length of the hose shall be not less than 15 feet. The hose shall be compatible for use with water, diesel fuel, and oil. The hose shall be stowed on the work platform or on brackets on the side of the tank provided the overall width of the vehicle is not increased to more than 96 inches.

* 3.8.4 Fire extinguisher compartment. A protected, open-faced compartment for a 10-pound dry chemical fire extinguisher, conforming to A-A-393, type II, class 2, size 10, shall be provided behind the right (curb side) fender. The interior of the compartment shall be painted a contrasting color. Quick-release type securing devices for the fire extinguisher shall be provided.

3.8.5 Ground cable reel. Unless otherwise specified (see 6.2), a ground cable reel shall be installed in a location that will permit an easy withdrawal and recovery of the grounding wire. The cable reel shall contain a ratchet for retaining the ground cable in any extended position so that tension is not maintained on the ground cable. The reel shall have an outlet guide. The wire shall be approximately 25 feet long. The wire shall be No. 12 stranded copper, or 0.094 inch diameter 7x7 stainless or galvanized steel, single conductor, petroleum resistant, rubber or nylon covered, plus Y ends 10 feet long. The Y ends shall be equipped with clamp connectors for grounding the trailer to ground and the trailer to the unit being serviced. The clamp connectors shall conform to MIL-C-83413, part number M83413. The free end of the cable shall be provided with an elastomeric bumper (ball). The ground cable reel shall be provided with an elastomeric bumper (ball). The ground cable reel shall be located on the right (curb) side. The ground cable reel mounting surfaces shall be free of paint and contamination, and shall be securely mounted to prevent the cable reel assembly from being pulled away from its mountings. A warning plate shall be attached to the cable reel stating: "WALK GROUNDING CABLE BACK TO THE REEL."

3.9 Plumbing system and engine driven pump. When specified (see 6.2), a plumbing system with an engine driven pump, together with associated hoses, faucets, valves and accessories, shall be furnished in addition to the equipment specified in 3.7 through 3.7.7 for type I and 3.8 through 3.8.5 for type II.

3.9.1 Plumbing system, type I. The plumbing system on type I shall provide for:

(a) Gravity discharge through the main shutoff valve and a gravity discharge faucet (3-inch) located at the front of the trailer. A cam lock fitting and a dust cap or dust plug, complete with security chain or cable, shall be provided for the 3-inch gravity discharge faucet.

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(b) Forced discharge through the main shutoff valve to a 1.500 inch faucet with cam lock fitting at the front of the trailer. A dust cap or dust plug and security chain or cable shall be provided for the 1.500 inch faucet.
(c) Self-loading of the tank from a nonpressurized external source (with the main shutoff valve closed, in through the 3 inch gravity discharge faucet by use of either 1.500 inch hose and the reducer specified in 3.9.19, through the pump, out through the 1.500 inch forced discharge faucet and into the tank through the manhole fill cover opening by use of the other 1.500 inch hose specified in 3.9.19). Bottom loading capability shall not be furnished.

3.9.2 Plumbing system, type II. The plumbing system on type II shall provide for:

(a) Gravity discharge through the emergency valve and the 3-inch gravity discharge faucet (see 3.8.2).
(b) Forced discharge through the emergency valve to a 1.500 inch faucet with cam lock fitting at the front of the trailer. A dust cap or dust plug and security chain or cable shall be provided for the 1.500 inch faucet.
(c) Self-loading of the tank from a nonpressurized external source (with the emergency valve closed, in through the 3-inch gravity discharge faucet by use of the 1.500 inch hose specified in 3.9.22, through the pump, out through the 1.500 inch forced discharge faucet and into the tank through the manhole fill cover opening by use of the 3-inch hose specified in 3.8.3). Bottom loading capability shall not be furnished.

3.9.3 Piping and couplings. All piping shall be not less than 3 inch inside diameter tubing with long sweep bends or 3 inch nominal diameter pipe with long sweep elbows. Piping shall be installed to withstand expansion and vibration without failure and shall conform to the strength requirements of DOT Specification DOT 406. All threads shall be of the American National Standard pipe thread series. Victaulic or equal couplings shall be used only on pipe (not on tubing).

3.9.4 Grooved end pipe clamp couplings or flanges. Clamp couplings or flanges, (Victaulic or equal) shall be provided on the inlet and outlet sides of the pump and strainer and on the faucets specified in 3.9.1 (type I) or 3.9.2 (type II).

3.9.5 Pipe grooves for couplings. Pipe grooves for couplings shall be in accordance with dimensions specified by the coupling manufacturer. The groove and outside diameters shall be concentric to each other. The groove and the ends of the pipe shall have a chamfer of 0.003 to 0.005 inch. The outside diameter of the pipe between the groove and the end of the pipe shall be smooth without evidence of scale, seams, weld splatter, digs or scratches. The fitting shall be

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capable of accepting a 100 psi air pressure test for at least 5 minutes without leakage. The fitting gasket shall be capable of service in minus 20°F to plus 180°F without evidence of deformation or change in dimensions. Gaskets on type I trailers shall conform to 3.1.1.4.

3.9.6 Pump. An engine driven, self-priming centrifugal pump shall be provided. Pump materials shall be corrosion resistant. The suction intake and pressure discharge ports of the pump shall be not less than 1.500 inches in diameter. A check valve shall be provided in the pump. A line strainer shall be furnished at the suction port of the pump. The line strainer shall be capable of removing any contamination, dirt and debris that would interfere with the pump operation. Drains in the system shall be provided to allow drainage of liquid(s) from the complete system. An instruction plate for the pump, the engine and the plumbing system operations shall be provided and shall be secured on the inside cabinet door.

3.9.7 Pump performance. The pump shall be capable of loading and unloading the tank body with liquid. The pump shall be capable of lifting the specified liquid(s) from 15 feet below ground level through the length of 1.500 inch hose specified in table III into the suction port of the pump. The rate of flow under these conditions shall be not less than the suction flow rate specified in table III. The pump shall be capable of discharging the specified liquid(s) from the tank, against the discharge head specified in table III, at the discharge flow rate specified in table III.

TABLE III. Pump performance requirements.

Suction			Discharge	
Trailer type	Length of hose	Flow rate (gpm)	Head ft. of water (pressure psi)	Flow rate (gpm)
I	25	40	81 (35)	35
II	100	30	104 (45)	100

* 3.9.8 Engine. An air cooled, four cycle, diesel engine shall be provided to drive the pump. The engine shall have a manual cranking system. The engine and pump shall be mounted to a common bedplate and shall be secured to the vehicle so as to minimize vibration. The engine shall have sufficient output power to drive the pump. During normal operating conditions, the continuous duty horsepower output required by the engine shall not exceed 75 percent of the full throttle rating of the engine.

3.9.9 Radio interference suppression. The trailer shall be suppressed to limit electromagnetic radiation in accordance with SAE J551.

3.9.10 Engine control panel. An engine accelerator, and key-operated start/stop switch shall be installed on a control panel in a weather and waterproof cabinet. Each control shall be identified as to function. The cabinet shall be furnished with a side hinged door and a spring loaded latch with a hasp for a 1-inch padlock.

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3.9.11 Fuel pump. A fuel pump shall be furnished on the engine. Gravity or syphon feed, as prohibited by Federal Motor Carrier Safety Regulation 393.65(d), shall not be furnished. The fuel system shall conform to Federal Motor Carrier Safety Regulation 393.65. When lifting fuel against a head of 4 feet, the pump shall have sufficient capacity to supply the engine at its maximum rated load. An in-line fuel filter shall be provided, either as a component of the fuel pump or between the fuel pump and the engine.

3.9.12 Governor. A variable-speed governor shall be furnished. The governor shall automatically control the engine speed at the selected rated speed at all fluid pressures and flows required. A manually-operated hand throttle shall control the engine at all speeds from the manufacturer's recommended minimum governed speed up to the maximum operating speed.

3.9.13 Engine fuel tank. An engine fuel tank conforming to Federal Motor Carrier Safety Regulation 393.67 shall be furnished. The fuel tank shall have a capacity to provide fuel for at least 4 hours of full-load operation. The fuel tank cap shall be painted red and shall be attached to the tank by means of a corrosion-resistant chain or cable. A sump suitable for the collection and accumulation of water shall be provided in the fuel tank. A drain shall be located at the bottom of the sump for complete drainage of the fuel tank. Drainage provisions shall be located to prevent spillage of fuel on any area of the vehicle. The tank filler shall be provided with a protective screen and a vent. A shut-off valve shall be provided in the fuel line to the engine adjacent to the fuel tank. A photo-etched corrosion-resistant plate shall be furnished and marked denoting the type of fuel to be used and shall be located in a conspicuous location near the filler cap.

* 3.9.14 Exhaust system. An exhaust system shall be furnished with a muffler. The muffler shall have a heat shield. The tail pipe shall be pointed toward the rear and to the ground. The exhaust system on type II trailers shall be furnished with a spark arrester. The spark arrester shall have an 80 percent efficiency rating when rated in accordance with SAE J350.

3.9.15 Equipment cabinet. An equipment cabinet shall be provided at the front of the trailer to house the plumbing system, pump and engine. The engine and exhaust system shall be isolated from the pump and the plumbing system by an insulated partition. The fuel tank filler pipe shall extend to the exterior of the equipment cabinet. The exhaust pipe inside the equipment cabinet shall be wrapped with fireproof non-asbestos insulation. Adequate ventilation shall be furnished. The equipment cabinet shall be of not less than 16 gage (0.0598 inch) stainless steel for type I and 16 gage sheet steel for type II, both reinforced double panel construction. Doors shall be vertically mounted and shall be weatherproof. The doors shall have continuous rod or hem type, heavy duty hinges, with stainless steel hinge pins or sockets. Latches shall have recessed handles. Cylinder type locks, keyed alike, shall be installed in each door. Moving parts of latches and

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locks shall be constructed of, or plated with, corrosion resistant material. Equipment cabinet doors over 44 inches in length shall have pillar bolts or two latches. Equipment cabinet doors shall have check arms to hold them in open position. For type I trailer, the equipment cabinet interior, including doors, shall be of a bright or satin stainless steel finish, unpainted. Welds shall be ground smooth. The interior shall be free of crevices.

3.9.16 Equipment cabinet floor. The cabinet floor on type I shall be of stainless steel with suitably located drain holes. The cabinet floor for type II shall be of not less than 3 pounds per square foot, flattened, expanded metal grating. The floor on both types shall be reinforced to accept the loads imposed by mounted equipment. All fasteners shall be corrosion-resistant.

3.9.17 Equipment cabinet lights. In addition to the lighting specified in 3.5.6, the following lighting and wiring requirements shall be furnished:

(a) A light shall be provided in both the engine section and in the plumbing system section of the equipment cabinet.
(b) On type II all wiring shall be in conduit with threaded joints, having waterproof and vaporproof connections at the lamps and junction boxes, but not to include boxes around the lights.

3.9.18 Unloading capability. The vents specified in 3.6.8 shall be supplemented, if necessary, to provide for forced discharge from the tank at the maximum capacity of the pump, with the manhole and fill cover sealed, without exceeding the vacuum limitations of DOT Safety Regulation 178.346, Specification DOT 406.

3.9.19 Hoses, type I. For type I trailer with a plumbing system and engine driven pump, two sections of non-collapsible ZZ-H-561 hose shall be provided, one section long enough to extend from the 1.500 inch pressure discharge faucet into the manhole fill cover, and one section not less than 25 feet long. The hoses shall be of not less than 1.500 inches in inside diameter. The hose shall be fitted with removable reducers, adapters and cam lock fittings as required to be compatible with both the 3 inch and 1.500 inch faucets specified in 3.9.1. The hose shall be capable of withstanding suction pressures of not less than 29 inches of mercury and discharge pressures of not less than 150 psi. Stowage provisions with retaining clamps shall be provided on the vehicle. An intake strainer for the hose end shall be provided as an accessory and stowed in the cabinet. The hoses and all associated fittings shall conform to 3.1.1.4.

3.9.20 Emergency valve front control, type II. The emergency valve (see 3.8.1) on type II trailers with a plumbing system and an engine driven pump shall be provided with a cable-linked, quick-acting opening and quick-acting closing control located in the equipment cabinet. The

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control shall be mounted to provide interference with one of the doors of the cabinet to insure that the emergency valve is closed when the cabinet doors are closed. The control shall have fusible links to insure automatic closing of the emergency valve in case of fire. One fusible link shall be located near the emergency valve and one shall be provided near the plumbing system.

3.9.21 Hose, type II. For type II trailer with a plumbing system and engine driven pump, a non-collapsible hose, suitable for water, diesel fuel and oil, shall be provided. The hose shall be of not less than 1.500 inches in inside diameter. The hose shall be fitted with removable reducers, adapters and cam lock fittings of the same size as and compatible with the faucets specified in 3.8.2 and 3.9.2(b). The hose shall be capable of withstanding suction pressures of not less than 29 inches of mercury and discharge pressures of not less than 150 psi. Stowage provisions with retaining clamps shall be provided on the vehicle. An intake strainer for the hose end shall be provided as an accessory and stowed in the cabinet. Unless otherwise specified, the hose shall be not less than 25 feet long. When specified (see 6.2), the hose shall be not less than 100 feet long.

3.9.22 Hose reel. When specified (see 6.2), a hose reel shall be furnished for the 1.500 inch diameter hose (both lengths of 1.500 inch hose for type I). The reel shall be of the manual rewind type. A brake or retarder shall be provided on the unwinding side and a crank shall be provided for rewinding the hose. Four-way rollers shall be provided to prevent damage to the hose.

3.10 Servicing and adjusting. Prior to the acceptance of the vehicle by the Government inspector, the contractor shall service and adjust each vehicle for operational use, including at least the following: alinement of wheels, inflation of all tires, and complete lubrication of chassis and running gear with grades of lubricants recommended for the ambient air temperature at the delivery point.

3.11 Workmanship. Defective components or parts and assemblies which have been repaired or modified to overcome deficiencies shall not be furnished. Welded, bolted, and riveted construction utilized shall be in accordance with the highest standards of the industry.

* 3.11.1 Welding. Welding procedures shall be in accordance with a nationally recognized welding code. The surface of parts to be welded shall be free from rust, scale, paint, grease, or other foreign matter. Welds shall be of sufficient size and shape to develop the full strength of the parts connected by the welds. Welds shall transmit stress without permanent deformation or failure when the parts connected by the weld are subjected to proof and service loadings. Interior tank welds shall be without peaks and craters.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements (examination and tests) as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

* 4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility for ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Government verification. Quality assurance operations performed by the contractor will be subject to Government verification at unscheduled intervals. Verification will consist of observations of the operations to determine that practices, methods, and procedures of the contractor's inspection are being properly applied. Failure of the contractor to promptly correct product deficiencies discovered shall be cause for suspension of acceptance until correction has been made or until conformance of product to specification criteria has been demonstrated.

4.3 First production vehicle inspection. The first vehicle of each type, size and model produced under the contract shall be inspected by the contractor at his plant under the direction and in the presence of Government representatives. The purpose of the inspection shall be to determine vehicle conformance to the contract. Acceptance of the first production vehicle shall not constitute a waiver by the Government of its rights under the provisions of the contract.

4.3.1 Vehicle weight. The first production vehicle shall be weighed to determine curb weight and the distribution of curb weight on the lunette and axles for model 2W trailer, and on the front and rear axles for model 4W trailer. The loads imposed on lunette and axles with the specified payload will be calculated to ascertain that the suspension, axles and tires furnished are of adequate capacity to meet contractual requirements.

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4.3.2 Leakage test. Each completely assembled tank body, including all piping, shall be tested for leaks at its design pressure in accordance with DOT Regulation 178.345. The tank of the first production vehicle shall be tested for leaks both before and after the road test.

* 4.3.3 Road test. The first production vehicle shall be road tested, with and without payload, by the contractor to assure the vehicle will operate in accordance with contractual requirements. Unless otherwise specified (see 6.2), the road test shall be for a distance of 100 miles. Half the mileage shall be accomplished with the trailer fully loaded and half shall be with the trailer empty. Approximately half the mileage shall be on paved streets or highways at speeds up to 55 mph and half shall be on gravel or dirt roads at speeds up to those deemed safe. At least five sudden stops shall be made from a speed of not less than 20 mph. Tracking ability shall be observed to verify conformance to 3.3.1; turning ability shall be observed to verify conformance to 3.3.2. After the test the trailer shall be examined for evidence of misalinement, binding or other malfunction. The test course shall be arranged so as to return the vehicle to the manufacturer's plant at the completion of the test for reinspection and chassis servicing. All costs of the test, including all costs for the towing vehicle, exclusive of personal expenses incurred by participating Government representatives, shall be borne by the contractor.

* 4.3.4 Test failure. The vehicle shall successfully complete the entire test. Failure of the vehicle to successfully complete the test shall be cause for nonacceptance of any of the contract quantity, pending correction of deficiencies and evidence of corrective action which shall prevent the recurrence of similar deficiencies in the contract quantity. Failure of the vehicle to successfully complete the tests shall not constitute an excusable delay in meeting scheduled deliveries. The equipment shall be closely monitored throughout the test for interference with towing operations, and for instability as evidenced by poor tracking or uncontrollable sideslipping, skidding, swerving, or tilting. The equipment also shall be closely monitored for loosening of parts, interference between parts, leakage of fluids or lubricants, overheating of components, and for damage, permanent distortion, or excessive wear of parts and components. The equipment shall be disassembled to the extent necessary for inspection for unusual wear or damage to components. Substitution of a new trailer or replacement of a major component may require complete retest at the discretion of the Government representatives. Rejection of the test trailer shall be for damage or deficiencies, including but not limited to the following:

(a) Damage caused by collision
(b) Failure of any major component
(c) Vibration due to misalinement of wheels or frame
(d) Vibration due to the type of construction or mounting
(e) Evidence of abnormal tire wear due to misalinement or unbalance

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(f) Failure of any vehicular safety device such as brakes or electrical circuits
(g) Evidence of structural weakness in any part of the vehicles, vehicle components, accessories or welds.
(h) Loose mountings of parts or accessories due to workmanship or vehicular operation.

4.3.5 Pump test. When an engine driven pump is specified, the pumping equipment shall be tested with water to determine conformance to the requirements specified in 3.9.7.

4.3.6 Breakaway test. On air hydraulic brake equipped vehicles, breakaway conditions shall be simulated by disconnecting brake supply lines and the brake system relay emergency valve shall provide and maintain brake application for not less than 15 minutes. At the end of the 15 minutes, air pressure bleed-down shall provide not less than a 5-pound per square inch (psi) drop in air pressure.

4.3.7 Emergency venting (type II). On type II trailer, the exposed surface area of the tank shall be measured and the capacity of each vent determined by its required stamping and certification (see DOT 178.345, 346). The total venting capacity shall be calculated and compared to the surface area to determine conformance to the requirements of DOT Safety Regulation 178.346.

4.3.8 Availability of regulations. The contractor shall make available to the Government at the point of final acceptance a current copy of DOT Safety Regulation 178.345 and DOT Safety Regulation 178.346. The documents shall be surrendered to the Government representatives for use in conjunction with the contract for the duration of the contract.

4.3.9 Joint specimens. The contractor shall offer for Government inspection and approval the joint test specimens subjected to the tensile test required by DOT Safety Regulation 178.345, together with certification regarding the date of preparation and testing. The test specimens shall be current in accordance with 178.345.

* 4.3.10 Toxic data and formulations. The supplier shall furnish a copy of the approval by the procuring activity of the materials used in the construction of the vehicle (see 3.1.1.4). The vehicle shall be inspected and the manufacturer shall provide evidence that only the approved materials were used in the construction and treatment of the affected parts of the vehicle.

4.3.11 Production sample. Upon acceptance of the first production vehicle, it shall remain at the manufacturing facility as a production sample, and shall be the last vehicle shipped on the contract. The contractor shall maintain the vehicle in a serviceable condition for the duration of the contract.

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4.3.12 Failure. Failure of the first production vehicle to meet requirements of the contract shall be cause for the Government to refuse acceptance of all vehicles under contract until corrective action has been taken.

4.4 Inspection of production vehicles. The contractor's inspection system shall, as a minimum, assure that the vehicle conforms to the physical and dimensional requirements and is capable of meeting performance requirements contained herein. All tank trailers under contract shall be tested for leaks as specified in 4.3.2. For each vehicle under contract, the contractor shall make available to the Government, at the point of final acceptance, records acceptable to the Government indicating that the servicing and adjusting required by 3.10 has been accomplished.

4.5 Certification. For each type II trailer, a manufacturer's certificate of compliance to DOT Specification DOT 406, as defined in 178.345, shall be furnished, endorsed by the vehicle contractor if different than the tank manufacturer, and securely stowed and shipped with the vehicle together with instructions for the recipient regarding the certificate's retention and use. The certificate shall bear, in addition to the signatures specified above, the typed and signed name of the Government official accepting the vehicle.

5. PACKAGING

5.1 Vehicle processing. The vehicle shall be processed for shipment, from the manufacturer's plant to the initial receiving activity, in accordance with the manufacturer's standard commercial practice.

6. NOTES

6.1 Intended use. The vehicles covered by this specification are intended for general nontactical use in transporting and dispensing: type I, potable water; type II, diesel fuel, oil and similar liquids, but not gasoline (see 6.3).

6.2 Ordering data. Acquisition documents should specify the following:

(a) Title, number, and date of this specification.
(b) Type, size and model of trailer required (see 1.2, 1.21 and 1.2.2).
(c) Issue of DODISS to be cited in the solicitation and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
(d) Identification of appropriate service for treatment, painting, marking and data plates (see 3.1.1.1 and 3.1.1.2).
(e) Rustproofing, if required (see 3.1.1.3).
(f) Low profile tires, if required (see 3.5.4.1).
(g) A spare wheel or rim, if required (see 3.5.4.3).
(h) A spare tire, if required (see 3.5.4.4).
(i) A spare tire carrier, if required (see 3.5.4.5).

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(j) An air brake synchronizing controller, if required in lieu of a hydraulic brake synchronizing controller (see 3.5.5.1).
(k) Air-hydraulic brakes, if required in lieu of electric brakes (see 3.4.5.2).
(1) Wheel chocks, if required (see 3.5.5.5).
(m) If an electrical jumper cable is not required (see 3.5.6.4).
(n) Pintle assembly, if required (see 3.5.9).
(o) If outer jacket and insulation are not required (see 3.7.7).
(p) Hand water pump, if required (see 3.7.8).
(q) If a fire extinguisher compartment is not required for type II (see 3.8.4).
(r) If a ground cable reel is not required for type II (see 3.8.5).
(s) Plumbing system and engine driven pump, if required (see 3.9).
(t) A 100 foot hose, if required in lieu of a 25 foot hose (see 3.9.21).
(u) Hose reel, if required (see 3.9.22).
(v) Road test mileage, if not 100 miles (see 4.3.3).

6.3 Gasoline. The type II trailers covered by this specification are not intended for transportation of gasoline. Federal and state vapor recovery requirements for vehicles transporting gasoline would cause a small 500 gallon vehicle to be equipped with many items that would far exceed in cost the benefits derived from the use of a 500 gallon trailer as opposed to larger vehicles (trucks and semitrailers).

6.4 Subject term (key word) listing.

Fuel tanker
Water tank.

6.5 Changes from previous issue. The margins of this document are marked with asterisks (*) to indicate where changes (additions, modifications, corrections and deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content regardless of the marginal notations and relationship to the previous issue.

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